SERVERLESS IMAGE/VIDEO PROCESSING

Services: S3, Lambda, Step Functions, Rekognition, SNS
Goal: Automatically process images or videos uploaded to S3 (e.g., resize, label detection).
Skills: Event-driven architecture, serverless orchestration.

Definition

This project implements a fully automated, event-driven serverless architecture on AWS that processes images uploaded to Amazon S3. When a new image is uploaded, an AWS Step Functions workflow is triggered automatically. The workflow orchestrates multiple AWS Lambda functions to perform image pre-processing, detect labels using Amazon Rekognition, and send notifications via Amazon SNS.
The architecture is fully serverless, scalable, and does not require any server management.

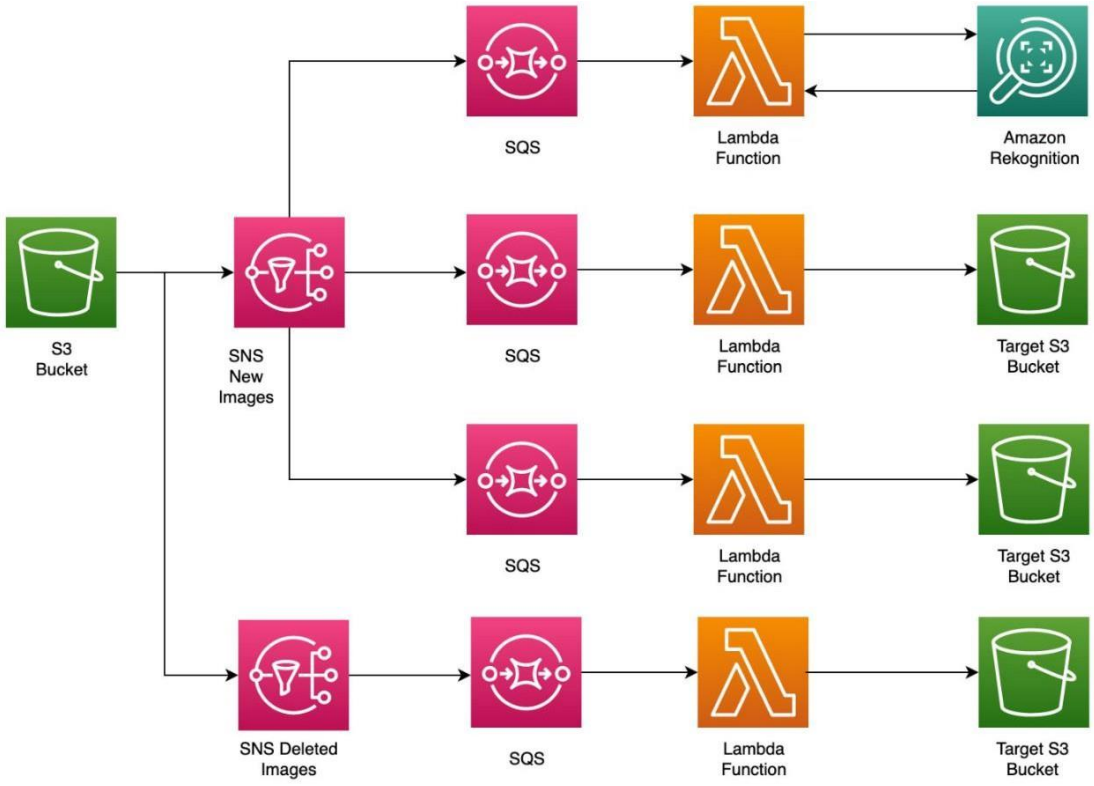
Scope of the Project

The scope of this project includes:
Automatic image processing on upload
AI-based label detection using Amazon Rekognition
Workflow orchestration using AWS Step Functions
Event-driven automation using S3 triggers
Email notification system using Amazon SNS
Implementation of secure IAM roles and permissions Future scope:
Video processing support
Face recognition integration
Error handling and retry mechanisms
Integration with API Gateway
Storage of results in DynamoDB
CI/CD automation for deployment

Methodology

The project follows an event-driven serverless architecture methodology.

Architecture Flow:



Key Design Principles:

Serverless computing (no EC2 usage)

Event-driven automation

Microservices-based Lambda functions

Workflow orchestration using Step Functions

Least privilege IAM security model

Modular and scalable architecture

Explanation of Services Used

◆ **Amazon S3**

Purpose:
Stores input images
Generates event notification on object upload
Acts as workflow trigger

◆ AWS Lambda

Purpose:
Executes backend logic without managing servers
ResizeImage Lambda: Pre-processing of image
DetectLabels Lambda: Calls Rekognition API
SendNotification Lambda: Publishes SNS notification
Trigger Lambda: Starts Step Function execution

◆ AWS Step Functions

Purpose:
Orchestrates multiple Lambda functions
Manages workflow sequence
Handles state transitions
Provides execution monitoring

◆ Amazon Rekognition

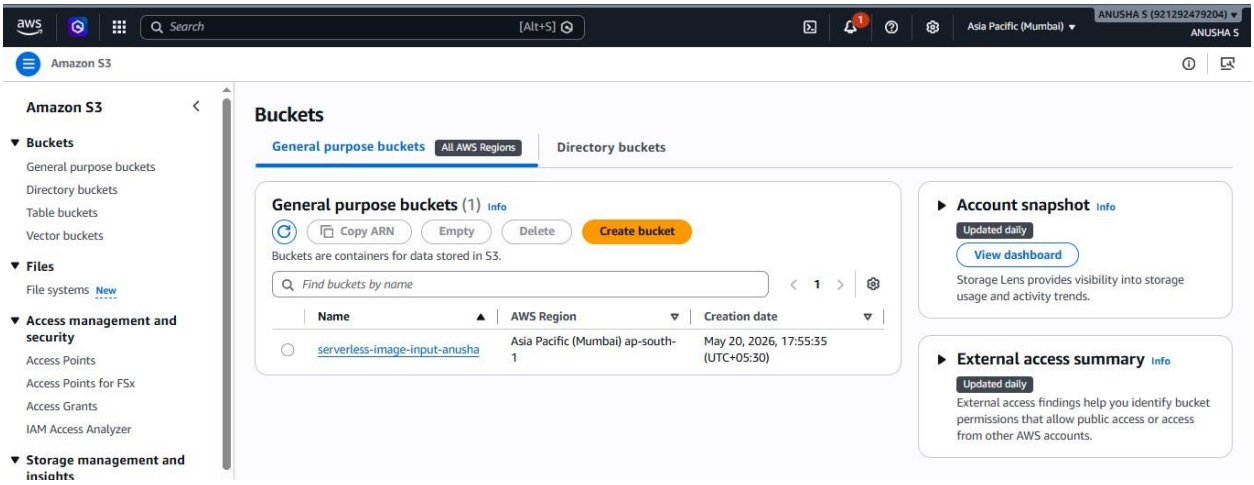
Purpose:
Performs AI-based label detection
Identifies objects within the image
Returns detected labels with confidence scores

◆ Amazon SNS
Purpose:
Sends email notifications
Publishes detected labels to subscribers

◆ IAM (Identity and Access Management)
Purpose:
Manages roles and permissions
Secures cross-service communication
Implements least privilege principle

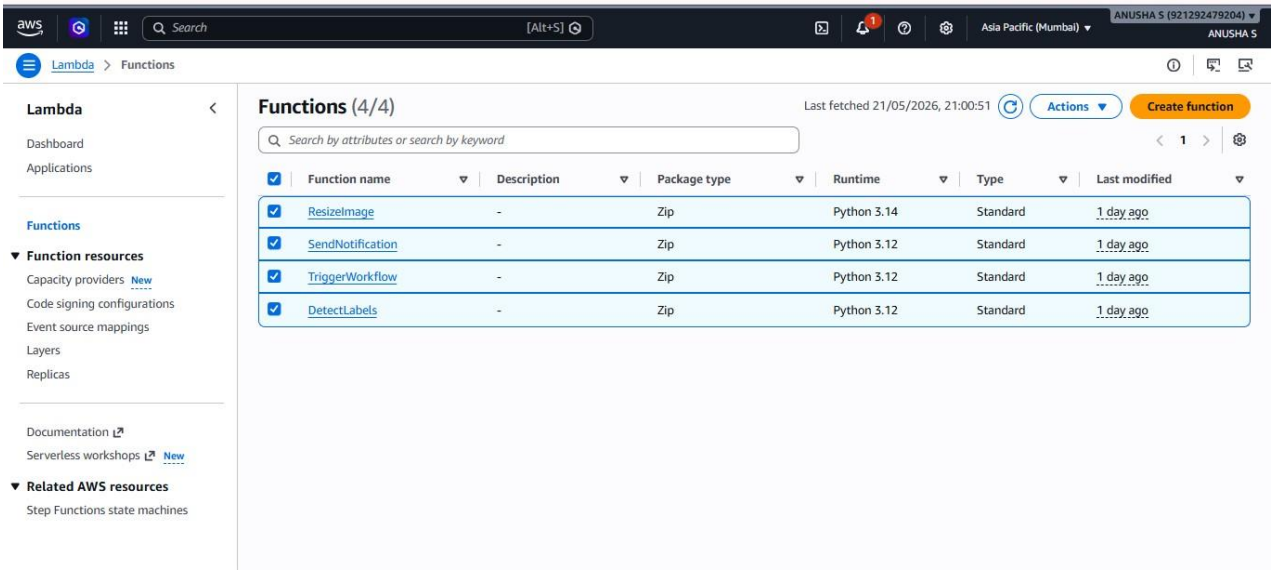
Step-by-Step Procedure

Step 1: Create S3 Bucket
Created input bucket: serverless-image-input-megha
Configured bucket for object creation events



Step 2: Create Lambda Functions
<u>ResizeImage</u> Lambda
Accepts bucket and key as input
Performs pre-processing

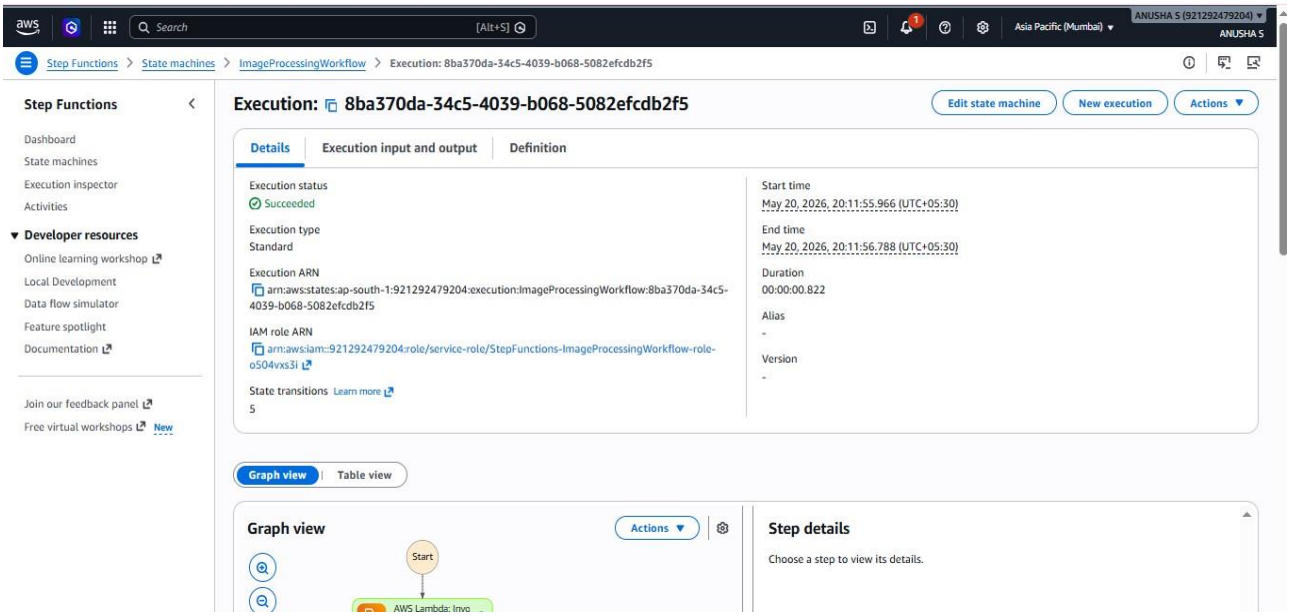
Returns bucket and key for next state
DetectLabels Lambda
Uses boto3 Rekognition client
Calls detect_labels API
Returns detected labels
SendNotification Lambda
Uses SNS client
Publishes label results to SNS topic
Trigger Lambda
Triggered by S3 event
Starts Step Function execution
Passes bucket and key as input

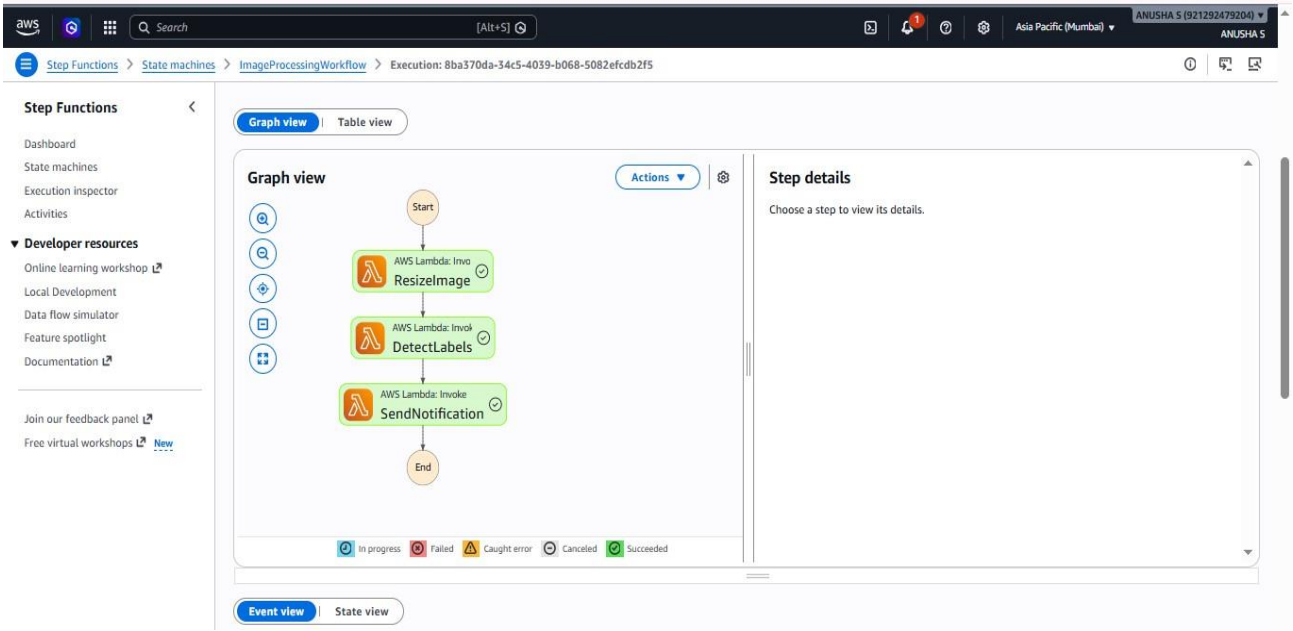


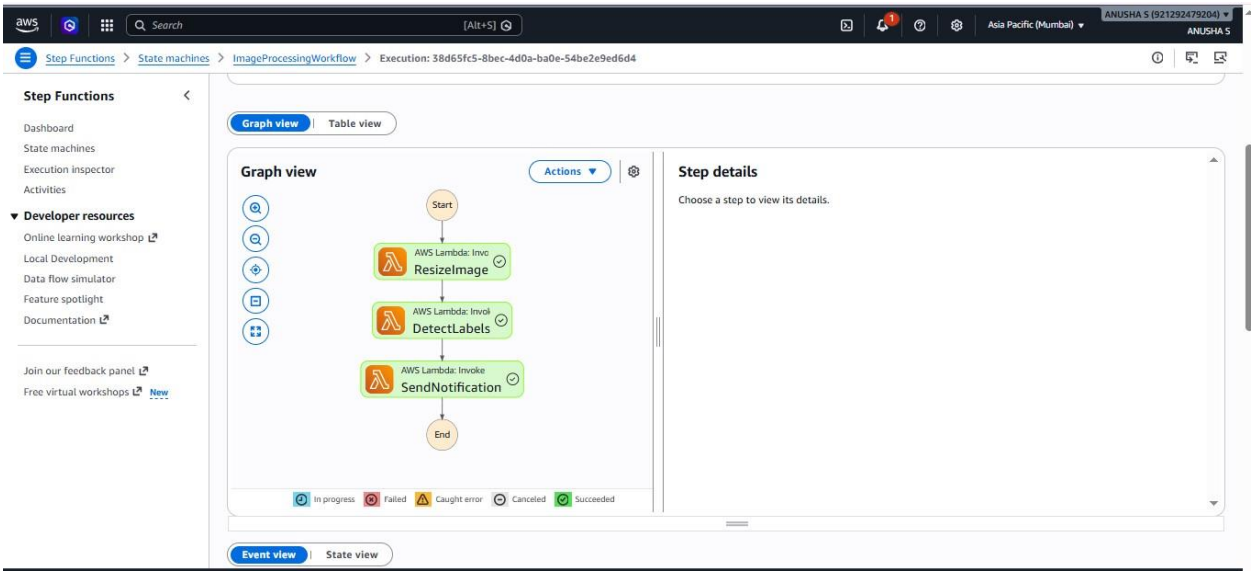
Step 3: Create SNS Topic
Created topic: image-processing-notification
Subscribed email endpoint
Confirmed email subscription

Step 4: Configure IAM Roles
Lambda execution roles created
Attached:
AWSLambdaBasicExecutionRole
AmazonS3ReadOnlyAccess
AWSStepFunctionsFullAccess
Added S3 invoke permission to trigger Lambda

Step 5: Create Step Function
Selected Standard workflow
Added states:
ResizeImage
DetectLabels
SendNotification
Configured JSONPath for input/output
Created execution role automatically

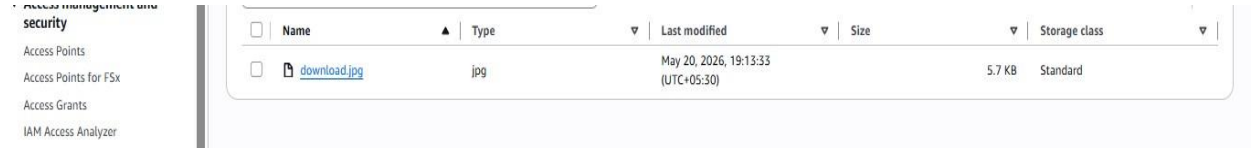

Step Functions > State machines > ImageProcessingWorkflow > Execution: 8ba370da-34c5-4039-b068-5082efcdb2f5
Execution: 8ba370da-34c5-4039-b068-5082efcdb2f5
Edit state machine
New execution
Actions
Details
Execution input and output
Definition
Execution status
Succeeded
Execution type
Standard
Execution ARN
arn:aws:states:ap-south-1:921292479204:execution:ImageProcessingWorkflow:8ba370da-34c5-4039-b068-5082efcdb2f5
IAM role ARN
arn:aws:iam::921292479204:role/service-role/StepFunctions-ImageProcessingWorkflow-role-o504vxs3i
State transitions
5
Start time
May 20, 2026, 20:11:55.966 (UTC+05:30)
End time
May 20, 2026, 20:11:56.788 (UTC+05:30)
Duration
00:00:00.822
Alias
Version
Graph view
Table view
Step details
Choose a step to view its details.


Step Functions > State machines > ImageProcessingWorkflow > Execution: 8ba370da-34c5-4039-b068-5082efcdb2f5
Graph view
Table view
Actions
Start
AWS Lambda: Invo
ResizeImage
AWS Lambda: Invok
DetectLabels
AWS Lambda: Invoke
SendNotification
End
In progress
Failed
Caught error
Canceled
Succeeded
Step details
Choose a step to view its details.
Event view
State view

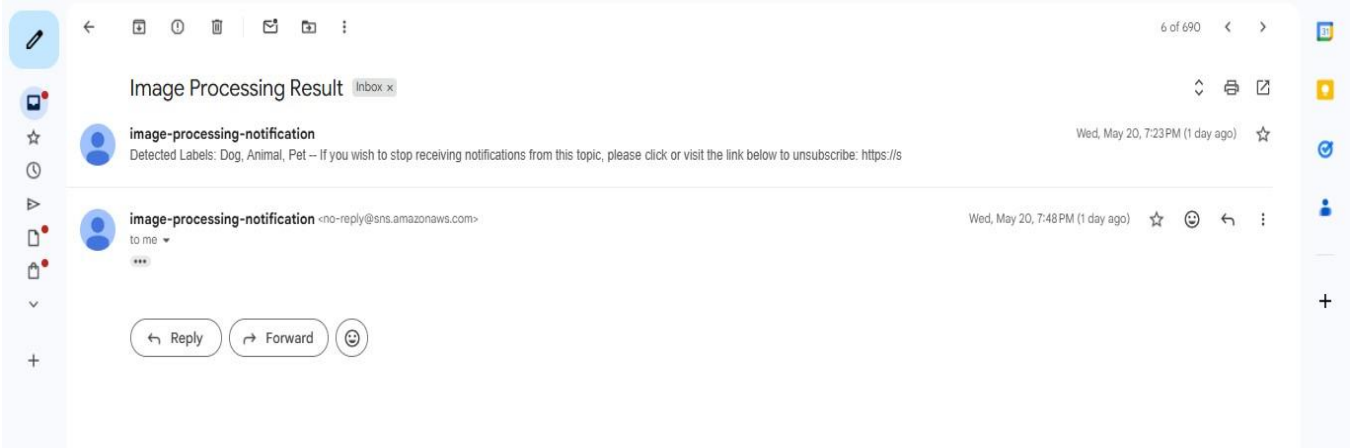


Step 6: Configure S3 Event Notification
Event type: All object create events
Destination: Trigger Lambda
Verified resource-based policy

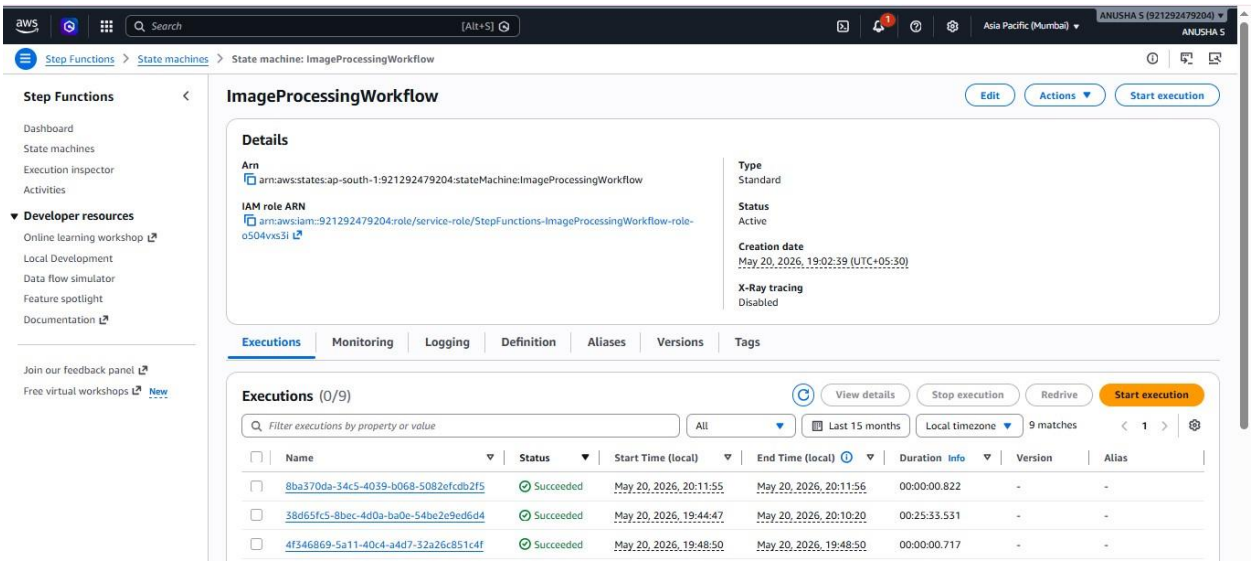
Step 7: Testing
Uploaded image to S3
Verified automatic execution



Checked logs in CloudWatch
Verified email notification



Confirmed successful execution



Troubleshooting

Issue 1: Execution Not Starting Automatically

Method: Checked Lambda logs Cause: S3 event notification missing

Solution: Created event notification properly

Issue 2: Log Group Not Created
Method: Checked CloudWatch Cause: Lambda never triggered
Solution: Verified S3 trigger configuration

Issue 3: InvalidS3ObjectException
Method: Checked DetectLabels step Cause: Missing S3 read permission
Solution: Added AmazonS3ReadOnlyAccess policy

Issue 4: Step Function Not Starting
Method: Checked Trigger Lambda logs Cause: Missing start_execution logic
Solution: Updated Lambda code and environment variable
Issue 5: InvalidClientTokenId (SNS)
Method: Checked region mismatch
Cause: SNS and Lambda in different region Solution: Created SNS in same region

Conclusion
This project successfully demonstrates the implementation of a fully automated serverless image processing workflow using AWS services. The architecture is event-driven, scalable, secure, and does not require infrastructure management.
Key Achievements:
Designed and implemented serverless architecture
Integrated AI services (Amazon Rekognition)
Implemented workflow orchestration using Step Functions
Configured IAM roles and cross-service permissions
Built automated notification system
Performed debugging and troubleshooting of real-world errors
The project showcases strong understanding of cloud-native architecture, automation, and DevOps practices.